Nikhil Narra

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Experience

Blackhawk Network **Apr. 2025 – Present**
Software Developer *Coppell, TX*

- Replaced a 3–4 hour nightly cron with an event-driven pipeline (AWS SQS, EventBridge) applying product updates to the commerce database in real time, eliminating 100K+ nightly product fetches; added client webhook notifications.
- Built an MDC-based distributed logging framework into the org's shared Java library, driving adoption across 15+ microservices and 5 backend platforms; correlation-ID tracing cut on-call log investigation from hours to minutes.
- Containerized legacy applications, including a Ruby on Rails storefront, migrating them from manually deployed EC2 to ECS behind ALBs with blue/green deployments, and stood up their first Jenkins CI/CD pipelines for PR builds and deploys.
- Built a dedicated AWS CERT environment mirroring pre-production across 15+ microservices and frontend/backend applications; provisioned infrastructure using YAML-based CloudFormation templates, and configured application environments.

Virginia Tech, Advanced Research Computing **Dec. 2023 – Apr. 2025**
Software Developer Intern *Blacksburg, VA*

- Built a RAG-based conversational search service over the university's public knowledge base using LangChain and LLMs, indexing 10K+ pages across 20+ departments for students and faculty to query via a chat web interface.
- Rebuilt legacy HTML coursework pages with embedded X3D simulations into a React platform with lazy-loading and performance optimizations, growing adoption from 3 to 40+ students with full iPad and mobile support.
- Contributed VR headset support to the open-source X3DOM library (200+ weekly npm downloads): integrated Meta Quest 3 controller models with fallback rendering for unrecognized headsets; tested releases and resolved open GitHub issues.

Deloitte **Jul. 2021 – Dec. 2022**
Analyst *Hyderabad, India*

- Developed Apex code and Salesforce Flows on the firm's internal employee-lifecycle platform, automating workflows spanning the full employment cycle from interview invitation through offboarding.
- Designed a UiPath RPA workflow with the RPA team to ingest scheduled employee-data feeds and update Salesforce records, eliminating 100% of a manual Data Loader upload process previously run 3–4 times daily on team rotation.
- Engineered a Salesforce-to-ServiceNow integration for the support-ticket migration: new Salesforce cases triggered templated emails parsed into replicated ServiceNow tickets via field-to-field mapping, preventing dropped cases.

Skills

Languages: Java, Python, JavaScript/TypeScript, SQL, C++
Backend: Spring Boot, Hibernate, Node.js, Express, Flask, REST APIs, Microservices, Event-Driven Architecture, JWT
Frontend: React, Angular
Cloud & DevOps: AWS, Docker, Jenkins, Azure DevOps, CI/CD, Git, Splunk, Elasticsearch/Kibana
Data & AI: PostgreSQL, MySQL, MongoDB, LangChain, RAG, LLM APIs (OpenAI), Model Context Protocol (MCP)

Projects

x3d-mcp – MCP Server for AI-Generated 3D Scenes | *Python, FastMCP, lxml*

- Built an open-source Model Context Protocol (MCP) server enabling LLMs to generate, edit, validate, and animate ISO-standard 3D scenes; merged into the **Web3D Consortium's official x3d_mcp** project, featuring 28 tools and four-level validation.

X3Test – Automated 3D Performance Benchmarking Framework | *Node.js, Puppeteer, CI/CD*

- Created an open-source npm library for headless benchmarking of X3D/X3DOM scenes (FPS, render time, node counts) via CLI and JavaScript API, built for CI/CD regression testing; won **3rd Place, Tool/Pipeline Innovation of the Year category** at the Web3D '25 Tools Competition.

Rural STARS – Gifted-Student Screening Platform | *Python, Streamlit, Pandas, ReportLab*

- Built a Streamlit platform scoring gifted-education screening surveys from webhook-fed QuestionPro responses in real time, normalizing Likert data and flagging candidates via per-scale thresholds across five talent domains.
- Automated school-facing reporting by engineering a validation pipeline that generates branded per-school PDF reports (ReportLab) with TLS-secured email delivery.

Publications & Honors

Scientific Excellence Award, ACM Web3D 2026 – x3d_mcp, recognized for advancing AI-assisted development
x3d_mcp: A Model Context Protocol Server for AI-Assisted X3D Authoring – *ACM Web3D '26* (DOI)
X3Test: Automated Performance Benchmarking of X3D/X3DOM Scenes – *ACM Web3D '25* (DOI)
A Generalized Web3D API for Metaverse Bookmarks – *ACM Web3D '25* (DOI)
Spot Award, Deloitte (2022) – recognized for technical ownership and high-impact delivery on enterprise automation initiatives

Education

Virginia Tech **Jan. 2023 – Dec. 2024**
Master of Science in Computer Science (GPA: 3.97 / 4.00) *Blacksburg, VA*

Jawaharlal Nehru Technological University, Hyderabad **Jul. 2017 – Jul. 2021**
Bachelor of Technology in Computer Science & Engineering *Hyderabad, India*